第七章 微扰理论

§7.1 不含时微扰

1. 非简并微扰

$\hat{H} = \hat{H}_0 + \hat{V}$

$\hat{H}_0$ 可以严格求解，$\hat{H}_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle$

将 $\hat{V}$ 视为微扰

$(\hat{H}_0 + \hat{V}) |n\rangle = (E_n^{(0)} + \Delta E_n) |n\rangle$

解为 $|n\rangle = \dfrac{\hat{V} - \Delta E_n}{E_n^{(0)} - \hat{H}_0} |n\rangle$ 为保证分母非0，$(\hat{V} - \Delta E_n)|n\rangle$ 不能包含 $|n\rangle$

改写 $= |n^{(0)}\rangle + \dfrac{1 - \hat{P}_n^{(0)}}{E_n^{(0)} - \hat{H}_0} (\hat{V} - \Delta E_n) |n\rangle$ $\quad \hat{P}_n^{(0)} = |n^{(0)}\rangle\langle n^{(0)}|$.

$\langle n^{(0)} | n\rangle = \langle n^{(0)} | n^{(0)}\rangle = 1$

利用了 $|n\rangle = |n^{(0)}\rangle + |n_s\rangle$

$= |n^{(0)}\rangle + (|n\rangle - |n^{(0)}\rangle) = |n^{(0)}\rangle + (|n\rangle - |n^{(0)}\rangle\langle n^{(0)} | n\rangle)$

$= |n^{(0)}\rangle + (1 - \hat{P}_n) |n\rangle$

$\langle n^{(0)} | \hat{H} + \hat{V} | n\rangle = \langle n^{(0)} | E_n^{(0)} + \Delta E_n | n\rangle$

$E_n^{(0)} + \langle n^{(0)} | \hat{V} | n\rangle = (E_n^{(0)} + \Delta E_n) \langle n^{(0)} | n\rangle$

$E_n^{(0)} + \langle n^{(0)} | \hat{V} | n\rangle = E_n^{(0)} + \Delta E_n$

能量修正值：$\Delta E_n = \langle n^{(0)} | \hat{V} | n\rangle$

将 ΔE_n，$|n\rangle$ 视为 $\hat{V}$ 的函数，展开

$\Delta E_n = \lambda E_n^{(1)} + \lambda^2 E_n^{(2)}$

$|n\rangle = |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle$

$\hat{H} = \hat{H}_0 + \lambda \hat{V}$

考察本征能量的一阶修正.

$\lambda E_n^{(1)} = \langle n^{(0)}|\hat V|n^{(0)}\rangle$　　$\lambda E_n^{(1)}$是1阶，$\hat V$是1阶，$|n^{(0)}\rangle$是0阶.

能量本征态：

$$|n^{(1)}\rangle = |n\rangle - |n^{(0)}\rangle = \frac{1-\hat P_n}{E_n^{(0)}-\hat H_0}(\hat V - \Delta E_n)|n\rangle$$

$\lambda = 1$

$$= \frac{1-\hat P_n}{E_n^{(0)}-\hat H_0}(\hat V - \lambda E_n^{(1)})\underline{|n^{(0)}\rangle} = \frac{1-\hat P_n}{E_n^{(0)}-\hat H_0}(\hat V - E_n^{(1)})|n^{(0)}\rangle$$

保留到一阶.

$$= \frac{1-\hat P_n}{E_n^{(0)}-\hat H_0}\hat V|n^{(0)}\rangle$$

$(1-\hat P_n)E_n^{(1)}|n^{(0)}\rangle = E_n^{(1)}(1-\hat P_n)|n^{(0)}\rangle = 0$

$$= \sum_{k\neq n}|k^{(0)}\rangle\frac{\langle k^{(0)}|\hat V|n^{(0)}\rangle}{E_n^{(0)}-E_k^{(0)}}$$

$1-\hat P_n = \sum_k |k\rangle\langle k| - |n\rangle\langle n|$

$$E_n^{(2)} = \langle n^{(0)}|\hat V|n^{(1)}\rangle = \sum_{k\neq n}\frac{|\langle k^{(0)}|\hat V|n^{(0)}\rangle|^2}{E_n^{(0)}-E_k^{(0)}}$$

$= \sum_{k\neq n}|k\rangle\langle k|$

$$|n^{(2)}\rangle = |n\rangle - |n^{(0)}\rangle - |n^{(1)}\rangle$$

$$= \frac{1-\hat P_n}{E_n^{(0)}-\hat H_0}\left[(\hat V - \Delta E_n)|n\rangle - \hat V|n^{(0)}\rangle\right]$$

$$= \frac{1-\hat P_n}{E_n^{(0)}-\hat H_0}\left[(\hat V - E_n^{(1)} - E_n^{(2)})(|n^{(0)}\rangle + |n^{(1)}\rangle) - \hat V|n^{(0)}\rangle\right]$$

$$= \frac{1-\hat P_n}{E_n^{(0)}-\hat H_0}\left[(\hat V - E_n^{(1)})|n^{(1)}\rangle - E_n^{(2)}|n^{(0)}\rangle - \underline{E_n^{(2)}|n^{(1)}\rangle}\right]$$

三阶，舍去.

$$= \frac{1-\hat P_n}{E_n^{(0)}-\hat H_0}(\hat V - E_n^{(1)})|n^{(1)}\rangle$$

$$= \sum_{m\neq n}\frac{|m\rangle\langle m|}{E_n^{(0)}-\hat H_0}(\hat V - \langle n^{(0)}|\hat V|n^{(0)}\rangle)\sum_{k\neq n}|k^{(0)}\rangle\frac{\langle k^{(0)}|\hat V|n^{(0)}\rangle}{E_n^{(0)}-E_k^{(0)}}$$

$$= \sum_{\substack{m\neq n\\ k\neq n}}|m^{(0)}\rangle\frac{\langle m^{(0)}|\hat V|k^{(0)}\rangle\langle k^{(0)}|\hat V|n^{(0)}\rangle}{(E_n^{(0)}-E_m^{(0)})(E_n^{(0)}-E_k^{(0)})} - \langle n^{(0)}|\hat V|n^{(0)}\rangle\cdot$$

$$\sum_{k\neq n}|k^{(0)}\rangle\frac{\langle k^{(0)}|\hat V|n^{(0)}\rangle}{E_n^{(0)}-E_k^{(0)}}$$

适用条件：$|\langle k^{(0)}|\hat V|n^{(0)}\rangle| \ll |E_n^{(0)} - E_k^{(0)}|$

例：将氢原子置于微弱电场中，求基态能量的变化。stark效应.

$\vec{\mathscr{E}} = \varepsilon\vec e_z$

$\hat V = e\vec r_1\cdot\vec{\mathscr{E}} - e\vec r_2\cdot\vec{\mathscr{E}} = e\vec r\cdot\vec{\mathscr{E}} = e\varepsilon\hat z$

基态能量修正：

一阶：$E_1^{(1)} = \langle 100|\hat V|100\rangle = e\varepsilon\langle 100|\hat z|100\rangle = 0$　?

二阶：$$E_2^{(1)} = \sum_{\langle\alpha lm\rangle\neq\langle 100\rangle}\frac{e^2\varepsilon^2|\langle\alpha lm|\hat V|100\rangle|^2}{E_1^{(0)}-E_\alpha^{(0)}}$$

$$= \sum \quad e^2\varepsilon^2|\langle\alpha 10|\hat V|100\rangle|^2 \quad = -\frac{9}{}\ \ a^3\varepsilon^2$$

$\alpha\neq 1$ $E_1^{(0)} - E_\alpha^{(0)}$

2. 简并微扰理论 · 未微扰的态存在简并

此时不满足 $\langle k^{(0)}|\hat{V}|n^{(0)}\rangle \ll |E_n^{(0)} - E_k^{(0)}|$

$\hat{P} = \sum_\alpha |n_\alpha^{(0)}\rangle\langle n_\alpha^{(0)}|$ ，$\{n_\alpha\}$ 为简并子空间内一组基

在简并子空间内， $\hat{H_0}|n_\alpha^{(0)}\rangle = E_n^{(0)}|n_\alpha^{(0)}\rangle$

$$(\hat{H_0} + \lambda\hat{V})|n\rangle = E_n|n\rangle$$

$\Leftrightarrow$

$$(\hat{H_0} + \lambda\hat{V})[\hat{P} + (1-\hat{P})]|n\rangle = E_n|n\rangle$$ 简并子空间内/外分开计算

① $\overset{\hat{P}}{\Leftrightarrow}$ $\hat{P}(\hat{H_0} + \lambda\hat{V})[\hat{P} + (1-\hat{P})]|n\rangle = E_n\hat{P}|n\rangle$

$$\hat{H_0}\hat{P}|n\rangle + \lambda\hat{P}\hat{V}\hat{P}|n\rangle + \lambda\hat{P}\hat{V}(1-\hat{P})|n\rangle = E_n\hat{P}|n\rangle \quad \hat{P}^2 = \hat{P}$$

$$(E_n - H_0 - \lambda\hat{P}\hat{V})\hat{P}|n\rangle = \lambda\hat{P}\hat{V}(1-\hat{P})|n\rangle \quad (A)$$

② $\overset{(1-\hat{P})}{\Leftrightarrow}$ $(1-\hat{P})[\hat{H_0} + \lambda\hat{V}][\hat{P} + (1-\hat{P})]|n\rangle = E_n(1-\hat{P})|n\rangle$

$$\hat{H_0}(1-\hat{P})|n\rangle + \lambda(1-\hat{P})\hat{V}\hat{P}|n\rangle + \lambda(1-\hat{P})\hat{V}(1-\hat{P})|n\rangle = E_n(1-\hat{P})|n\rangle$$

$$[E_n - \hat{H_0} - \lambda(1-\hat{P})\hat{V}](1-\hat{P})|n\rangle = \lambda(1-\hat{P})\hat{V}\hat{P}|n\rangle$$

$$\Leftrightarrow (1-\hat{P})|n\rangle = \frac{1}{E_n - \hat{H_0} - \lambda(1-\hat{P})\hat{V}}\lambda(1-\hat{P})\hat{V}\hat{P}|n\rangle$$

$$(1-\hat{P})^2|n\rangle = (1-\hat{P})^2\frac{1}{(1-\hat{P})(E_n - \hat{H_0} - \lambda\hat{V})}\lambda(1-\hat{P})\hat{V}\hat{P}|n\rangle$$

$$(1-\hat{P})|n\rangle = (1-\hat{P})\frac{1}{E_n - \hat{H_0} - \lambda\hat{V}}\lambda(1-\hat{P})\hat{V}\hat{P}|n\rangle \quad (B)$$

(A)，(B)：

$$(E_n - H_0 - \lambda\hat{P}\hat{V})\hat{P}|n\rangle = \lambda\hat{P}\hat{V}(1-\hat{P})\frac{1}{E_n - \hat{H_0} - \lambda\hat{V}}\lambda(1-\hat{P})\hat{V}\hat{P}|n\rangle$$

$$\left\{\hat{H_0} + \lambda\hat{P}\hat{V} + \lambda^2\hat{P}\hat{V}(1-\hat{P})\frac{1}{E_n - \hat{H_0} - \lambda\hat{V}}(1-\hat{P})\hat{V}\right\}\hat{P}|n\rangle = E_n\hat{P}|n\rangle$$

$$\hat{P}\left[H_0 + \lambda\hat{V} + \lambda^2\hat{V}(1-\hat{P})\frac{1}{E_n - \hat{H_0} - \lambda\hat{V}}(1-\hat{P})\hat{V}\right]\hat{P}|n\rangle = E_n\hat{P}|n\rangle$$

定义简并子空间内有效哈密顿量

$$\hat{H}_{eff} = \hat{P}\left[H_0 + \lambda\hat{V} + \lambda^2\hat{V}(1-\hat{P})\frac{1}{E_n - \hat{H_0} - \lambda\hat{V}}(1-\hat{P})\hat{V}\right]\hat{P}$$

本征方程： $\hat{H}_{eff}\hat{P}|n\rangle = E_n\hat{P}|n\rangle$

$$\hat{H}_{eff}|n^{(0)}\rangle = E_n|n^{(0)}\rangle$$

将本征能量展开：

$$E_n = E_n^{(0)} + \Delta E_n = E_n^{(0)} + \lambda E_n^{(1)} + \cdots$$

$$(1-\hat{P})\frac{1}{E_n - \hat{H}_0 - \lambda\hat{V}}(1-\hat{P}) = (1-\hat{P})\frac{1}{E_n^{(0)} - \hat{H}_0 - (\lambda\hat{V} - \Delta E_n)}(1-\hat{P})$$

$$= \frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\sum_{k=0}\left[(\lambda\hat{V} - \Delta E_n)\frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\right]^k$$

代回得到

$$\hat{H}_{eff} = \hat{P}\left[H_0 + \lambda\hat{V} + \lambda^2\hat{V}\frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\sum_{k=0}\left[(\lambda\hat{V} - \Delta E_n)\frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\right]^k\hat{V}\right]\hat{P}$$

$$|n\rangle = |n^{(0)}\rangle + (1-\hat{P})|n\rangle$$

$$= |n^{(0)}\rangle + (1-\hat{P})\frac{1}{E_n - \hat{H}_0 - \lambda\hat{V}}\lambda(1-\hat{P})\hat{V}\hat{P}|n\rangle \quad (B)$$

$$= |n^{(0)}\rangle + \frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\sum_{k=0}\left[(\lambda\hat{V} - \Delta E_n)\frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\right]^k\lambda\hat{V}\hat{P}|n\rangle$$

$$= |n^{(0)}\rangle + \frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\sum_{k=0}\left[(\lambda\hat{V} - \Delta E_n)\frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\right]^k\lambda\hat{V}|n^{(0)}\rangle$$

一级近似， $\hat{H}_{eff} = \hat{P}(\hat{H}_0 + \hat{V})\hat{P}$　$\lambda = 1$

本征方程： $\hat{P}\hat{V}\hat{P}|n^{(0)}\rangle = E_n^{(1)}|n^{(0)}\rangle$

设第n个能级为g重简并． $|n^{(0)}\rangle = \sum_{\alpha=1}^{g} C_\alpha^{(0)}|n_\alpha^{(0)}\rangle$ 在简并子空间里展开.

$$\begin{pmatrix} V_{11} & \cdots & V_{1g} \\ \vdots & & \\ V_{g1} & \cdots & V_{gg} \end{pmatrix}\begin{pmatrix} C_{n1}^{(0)} \\ \vdots \\ C_{ng}^{(0)} \end{pmatrix} = E_n^{(1)}\begin{pmatrix} C_{n1}^{(0)} \\ \vdots \\ C_{ng}^{(0)} \end{pmatrix}$$ 其中 $V_{\alpha\beta} = \langle n_\alpha^{(0)}|\hat{V}|n_\beta^{(0)}\rangle$

此时 $|n\rangle = |n^{(0)}\rangle + \frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\hat{V}|n^{(0)}\rangle = \left(1 + \frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\hat{V}\right)|n^{(0)}\rangle$　(只保留 $k=0$ 项)

$$E_n^{(1)} = \langle n^{(0)}|\hat{P}\hat{V}\hat{P}|n^{(0)}\rangle$$

二级近似：

$$\hat{H}_{eff} = \hat{P}\left(\hat{H}_0 + \hat{V} + \hat{V}\frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\hat{V}\right)\hat{P}$$

本征方程： $\hat{P}\left(\hat{V} + \hat{V}\frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\hat{V}\right)\hat{P}|n^{(0)}\rangle = (E_n^{(1)} + E_n^{(2)})|n^{(0)}\rangle$

本征态： $|n\rangle = |n^{(0)}\rangle + \frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\left[1 + (\hat{V} - \Delta E_n)\frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\right]\hat{V}|n^{(0)}\rangle$

$$= \left[1 + \frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\hat{V} + \left(\frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\hat{V}\right)^2 - \left(\frac{1-\hat{P}}{E_n^{(0)} - \hat{H}_0}\right)^2\hat{V}E_n^{(1)}\right]|n^{(0)}\rangle$$

能量： $E_n^{(2)} = \langle n^{(0)} | \hat{P}\hat{V} \frac{1-\hat{P}}{E_n^{(0)}-\hat{H}_0} \hat{V}\hat{P} | n^{(0)} \rangle$

※ 未知板书

定义： $\hat{G}_0 = \frac{1}{E_n^{(0)} - (1-\hat{P})\hat{H}_0(1-\hat{P})} = \frac{1-\hat{P}}{E_n^{(0)}-\hat{H}_0}$

$$\hat{G} = \frac{}{E_n - (1-\hat{P})\hat{H}_0(1-\hat{P}) - (1-\hat{P})\hat{V}(1-\hat{P})}$$

$$\hat{G}^{-1} = \hat{G}_0^{-1} - [(1-\hat{P})\lambda\hat{V}(1-\hat{P}) - \Delta E_n]$$

$\hat{G}_0$ 口 $\hat{G}$

$$\hat{G} = \hat{G}_0 + \hat{G}_0[(1-\hat{P})\lambda\hat{V}(1-\hat{P}) - \Delta E_n]\hat{G}$$

例：将氢原子置于微弱静电场中，求第一激发态的能量修正.

$|200\rangle, |210\rangle, |211\rangle, |21-1\rangle$

$\hat{V} = e\mathcal{E}\hat{z}$

$\Delta l = \pm 1 \to V_{\alpha\beta} \neq 0$ $\quad [\hat{z}, \hat{L}_z] = 0$

$\langle 200|\hat{V}|210\rangle = \alpha, \quad \langle 210|\hat{V}|200\rangle = -\alpha$

例：考虑固体中两个磁性离子，

§ 7.2 含时微扰

1. 一般形式

$$\hat{H} = \hat{H}_0 + \hat{V}(t)$$

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = \hat{H} |\psi\rangle$$

$$|\psi\rangle = \sum_n C_n(t)\, e^{-\frac{iE_n t}{\hbar}} |n\rangle$$

$$\Rightarrow \left(i\hbar \frac{\partial}{\partial t} C_n(t) + C_n E_n\right) e^{-\frac{iE_n t}{\hbar}} |n\rangle = \sum_n C_n(t)\, e^{-\frac{iE_n t}{\hbar}} (E_n + \hat{V}(t)) |n\rangle$$

$$\frac{\partial}{\partial t} C_m(t) = \frac{1}{i\hbar} \sum_n C_n(t)\, e^{i\frac{E_m - E_n}{\hbar} t} \langle m | \hat{V} | n \rangle$$

$$C_m(t) = C_m(0) + \int_0^t \frac{1}{i\hbar} \sum_n C_n(t')\, e^{i\frac{E_m - E_n}{\hbar} t'} \langle m | \hat{V} | n \rangle\, dt'$$

$$迭代 = C_m(0) + \int_0^t \frac{1}{i\hbar} \sum_n C_n(0)\, e^{i\frac{E_m - E_n}{\hbar} t'} \langle m | \hat{V} | n \rangle\, dt'$$

$$+ \frac{1}{(i\hbar)^2} \sum_{nk} \int_0^t dt' \int_0^t dt'' \, C_n(0)\, e^{i\frac{E_m - E_k}{\hbar} t'} e^{i\frac{E_k - E_n}{\hbar} t''} \langle m | \hat{V}(t') | k \rangle \langle k | \hat{V}(t'') | n \rangle$$

考虑一级近似：

$$C_m(t) = C_m(0) + \int_0^t \frac{1}{i\hbar} \sum_n C_n(0)\, e^{i\frac{E_m - E_n}{\hbar} t'} \langle m | \hat{V} | n \rangle\, dt'$$

设 $t=0$ 时，系统处在 H_0 本征态 $|i\rangle$

$$C_n(0) = \delta_{ni}$$

$\forall f \neq i$：

$$C_{fi}^{(1)}(t) = C_f(t) = \frac{1}{i\hbar} \int_0^t dt'\, e^{\frac{i(E_f - E_i)t}{\hbar}} \langle f | \hat{V}(t) | i \rangle$$

经过 t 时间后，系统跃迁到 $|f\rangle$ 的概率.

$$P_{fi}(t) = |C_{fi}(t)|^2 = \frac{1}{\hbar^2} \left| \int_0^t dt'\, e^{\frac{i(E_f - E_i)t}{\hbar}} \langle f | \hat{V}(t) | i \rangle \right|$$

2、简谐振荡微扰

$$\hat{V}(t) = 2\hat{V}\cos\omega t$$

$$C^{(1)}_{fi}(t) = \int_0^t dt' \langle f|\hat{V}|i\rangle\, 2\cos\omega t'\, e^{i\omega_{fi}t'}$$

$$= -\frac{V_{fi}}{\hbar}\left|\frac{e^{i(\omega_{fi}+\omega)t}-1}{\omega_{fi}+\omega} + \frac{e^{i(\omega_{fi}-\omega)t}-1}{\omega_{fi}+\omega}\right|$$

$$\omega_{fi} = \frac{E_f - E_i}{\hbar}$$

$$V_{fi} = \langle f|\hat{V}|i\rangle$$

$\omega \approx \omega_{fi}$ 或 $\omega \approx -\omega_{fi}$ 时，跃迁概率增加.

对应系统吸收/释放了 $\hbar\omega$ 的能量后，$|i\rangle \to |f\rangle$

$$P_{fi}(t) = |C_{fi}(t)|^2 = \frac{V_{fi}^2}{\hbar^2}\left|\frac{e^{i(\omega_{fi}+\omega)t}-1}{\omega_{fi}+\omega} + \frac{e^{i(\omega_{fi}-\omega)t}-1}{\omega_{fi}+\omega}\right|^2$$

$$= \frac{4|V_{fi}|^2}{\hbar^2(\omega_f-\omega_i)^2}\sin^2\frac{1}{2}(\omega_{fi}-\omega)t$$

$t \to \infty$ 有， $\frac{\sin^2\alpha x}{x^2} \xrightarrow{\alpha\to\infty} \pi\alpha\,\delta(x)$

$$P_{fi}(t) = \frac{2\pi|V_{fi}|^2 t}{\hbar}\delta(E_f - E_i - \hbar\omega)$$

跃迁速率： $w_{fi}(t) = \frac{dP_{fi}(t)}{dt}$

$$= \frac{2\pi|V_{fi}|^2}{\hbar}\delta(E_f - E_i - \hbar\omega)$$

Fermi黄金规则：

$$w_{fi}(t) = \int_{E_f-\frac{\Delta E}{2}}^{E_f+\frac{\Delta E}{2}} dE\, \rho_f(E)\, \frac{2\pi|V_{fi}|^2}{\hbar}\delta(E-E_i-\hbar\omega)$$

$$= \frac{2\pi|V_{fi}|^2}{\hbar}\rho_f(E_i+\hbar\omega)$$

3. 应用：原子对电磁波的吸收.

$$\hat{H} = \frac{1}{2m}(\vec{p}+e\hat{A})^2 - e\hat{\Phi} + \hat{V}(\vec{r})$$

若电磁波波长 λ>> 原子半径 a，近似认为原子附近电磁场均匀.

$$\hat{A} = \hat{A}(t) \approx \hat{A}_0\cos(\omega t)$$

做规范变换 $\hat{G} = e^{ie\frac{\hat{r}\cdot\hat{A}}{\hbar}}$

$$\begin{cases} \hat{A}'(t) = \hat{A}(t) + \nabla\left[-\hat{r}\cdot\hat{A}(t)\right] = 0 \\ \hat{\Phi}' = 0 - \frac{\partial}{\partial t}\left[-\hat{r}\cdot\hat{A}(t)\right] = \hat{r}\cdot\frac{\partial\hat{A}(t)}{\partial t} \end{cases}$$

$$\hat{H}' = \frac{\hat{p}^2}{2m} + \hat{V}(\hat{r}) - \frac{\partial\hat{A}(t)}{\partial t}\cdot e\hat{r}$$ $\hat{E}(t) = -\nabla\hat{\Phi}' - \frac{\partial\hat{A}'}{\partial t} = -\frac{\partial}{\partial t}\hat{A}(t)$

$$= \frac{\hat{p}^2}{2m} + \hat{V}(\hat{r}) + \underline{\hat{E}(t)\cdot e\hat{r}}$$ 微扰项 电偶极近似.

$$= \hat{H}_0' + \hat{H}_1'$$

$$C_{fi}^{(1)}(t) = -\frac{\langle f|\hat{E}_0\cdot e\vec{r}|i\rangle}{2\hbar}\left[\frac{e^{i(\omega_{fi}+\omega)t}-1}{\omega_{fi}+\omega} + \frac{e^{i(\omega_{fi}-\omega)t}-1}{\omega_{fi}+\omega}\right]$$ $\omega_{fi} = \frac{E_f - E_i}{\hbar}$

$$P_{fi}(t)\Big|_{t\to\infty} = \frac{\pi t}{2\hbar^2}\left|\langle f|\hat{E}_0\cdot e\vec{r}|i\rangle\right|^2\delta(\omega_{fi}-\omega)$$

$$w_{fi}(t) = \frac{dP_{fi}(t)}{dt} = \frac{\pi}{2\hbar^2}\left|\langle f|\hat{E}_0\cdot e\vec{r}|i\rangle\right|^2\delta(E_f - E_i - \hbar\omega)$$

选择定则：

$$\langle nlm|\vec{r}|n'l'm'\rangle \neq 0 \Rightarrow \Delta l = \pm 1$$

(1) 发射/吸收z方向的偏振光

$$\langle nlm|[\hat{L}_z, \hat{z}]|n'l'm'\rangle = 0$$

$$\Rightarrow (m'-m)\hbar\langle nlm|\hat{z}|n'l'm'\rangle = 0,$$

$$\Delta m = 0$$

(2) x, y 方向

$$\begin{cases} \langle n'l'm'|[\hat{L}_z, \hat{x}]|nlm\rangle = (m'-m)\hbar\langle n'l'm'|\hat{x}|nlm\rangle = i\hbar\langle n'l'm'|\hat{y}|nlm\rangle \\ (m'-m)\hbar\langle n'l'm'|\hat{y}|nlm\rangle = -i\hbar\langle n'l'm'|\hat{x}|nlm\rangle \end{cases}$$

$$\Rightarrow \Delta m = \pm 1$$

作用：① 跃迁

② 受激辐射

③ 自发辐射.

作业.

3. $\hat{H} = \alpha \hat{S}_z + \beta(3\hat{S}_x^2 + \hat{S}_y^2)$

$S=2$ 在 S_z 表象，$\{|2\rangle, |-2\rangle, |0\rangle, |1\rangle, |-1\rangle\}$

$\hat{H}_0 = \alpha\hbar^2 \,\mathrm{diag}(4, 4, 0, 1, 1)$

$$\hat{H}' = \beta\hbar^2 \begin{pmatrix} 4 & 0 & \sqrt{6} & & \\ 0 & 4 & \sqrt{6} & & 0 \\ \sqrt{6} & \sqrt{6} & 12 & & \\ & 0 & & 10 & 3 \\ & & & 3 & 10 \end{pmatrix}$$

H_0 : $E_1 = E_2 = 4\alpha^2\hbar^2$, $E_3 = 0$, $E_4 = E_5 = \alpha\hbar^2$

二级近似下的微扰矩阵.

$$\begin{pmatrix} H'_{11} + \end{pmatrix}$$

$$= \beta\hbar^2 \begin{pmatrix} 4 + \frac{3\beta}{2\alpha} & \frac{3\beta}{2\alpha} \\ \frac{3\beta}{2\alpha} & 4 + \frac{3\beta}{2\alpha} \end{pmatrix}$$

$E_1^{(1)} + E_1^{(2)} = 4\beta\hbar^2$

$E_2^{(1)} + E_2^{(2)} = 4\beta\hbar^2 + \frac{3\beta\hbar^2}{\alpha}$

第3个：

$E_3^{(0)} = 0$,

$E_3^{(1)} = \langle 0 | H' | 0 \rangle = 12\beta\hbar^2$

$$E_3^{(2)} = \frac{H'^2_{13}}{E_3^{(0)} - E_1^{(0)}} + \frac{H'^2_{23}}{E_3^{(0)} - E_2^{(0)}} = -\frac{3\beta\hbar^2}{\alpha}$$

第4, 5能级：

微扰矩阵 $\beta\hbar^2 \begin{pmatrix} 10 & 3 \\ 3 & 10 \end{pmatrix}$

$E_4^{(1)} = 7\beta\hbar^2$

$E_5^{(1)} = 13\beta\hbar^2$

5. 原：$E_n = (n+\frac{1}{2})\hbar\omega$, $|n\rangle$

$$\hat{H}' = q\varepsilon_0 \hat{x} e^{-t/\tau}$$

$$\langle i|\hat{H}'|j\rangle = q\varepsilon_0 \langle i|\hat{x}|j\rangle e^{-t/\tau}$$

$$= q\varepsilon_0 \sqrt{\frac{\hbar^2}{2m\omega}} \langle i|\hat{a}+\hat{a}^+|j\rangle e^{-t/\tau}$$

$\Rightarrow$ 选择定则：$\Delta n = \pm 1$

(2)

跃迁概率 $P_{10} = \frac{1}{\hbar^2}\left|\int_0^\infty \langle 1|\hat{H}'(t)|0\rangle e^{i\frac{E_0-E_1}{\hbar}t} dt\right|^2$

$$= \frac{q^2\varepsilon_0^2\tau^2}{2\hbar^2 m\omega(1+\omega^2\tau^2)}$$

留下概率 $P_0 = 1 - P_{10}$

3. $V_a(x) = \lambda x^4$

$\hat{U}(x) = V + V_a$

$|n'\rangle$

基态：包含原谐振子 $n = 0, 2, 4$.

可跃迁到 $n = 1, 3, 5$.

$n' = 1, 3, 5, 7, 9$.